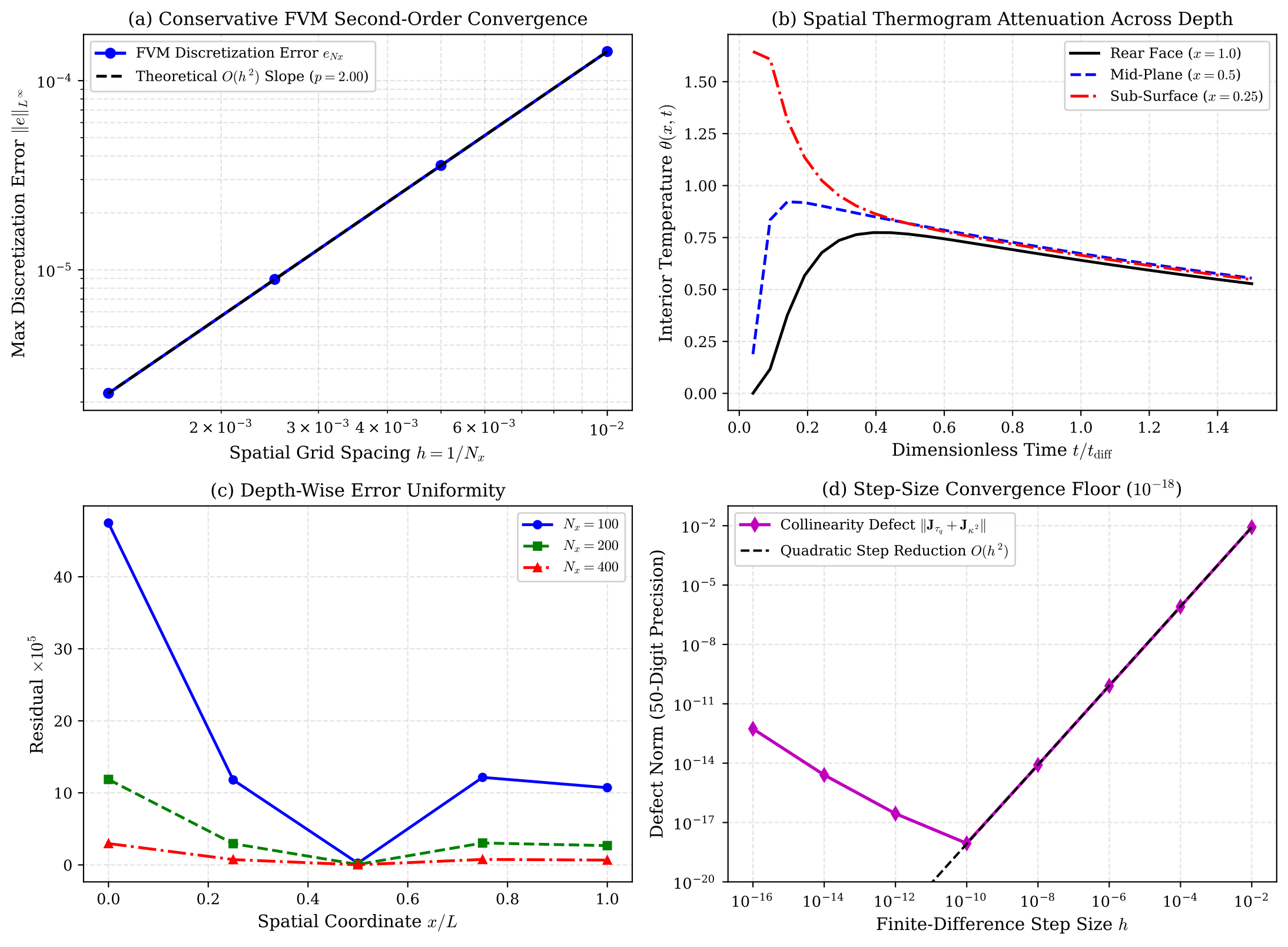
(a) Conservative FVM Second-Order Convergence
FVM Discretization Error e_{Nx}
Theoretical $O(h^2)$ Slope ($p = 2.00$)
Max Discretization Error $\|e\|_{L^\infty}$
10^{-4}
10^{-5}
2×10^{-3}
3×10^{-3}
4×10^{-3}
6×10^{-3}
10^{-2}
Spatial Grid Spacing $h = 1/N_x$
(b) Spatial Thermogram Attenuation Across Depth
Rear Face ($x = 1.0$)
Mid-Plane ($x = 0.5$)
Sub-Surface ($x = 0.25$)
Interior Temperature $\theta(x, t)$
0.00
0.25
0.50
0.75
1.00
1.25
1.50
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Dimensionless Time t/t_{diff}
(c) Depth-Wise Error Uniformity
$N_x = 100$
$N_x = 200$
$N_x = 400$
Residual $\times 10^5$
0
10
20
30
40
0.0
0.2
0.4
0.6
0.8
1.0
Spatial Coordinate x/L
(d) Step-Size Convergence Floor (10^{-18})
Collinearity Defect $\|\mathbf{J}_{\tau_q} + \mathbf{J}_{\kappa^2}\|$
Quadratic Step Reduction $O(h^2)$
Defect Norm (50-Digit Precision)
10^{-2}
10^{-5}
10^{-8}
10^{-11}
10^{-14}
10^{-17}
10^{-20}
10^{-16}
10^{-14}
10^{-12}
10^{-10}
10^{-8}
10^{-6}
10^{-4}
10^{-2}
Finite-Difference Step Size h